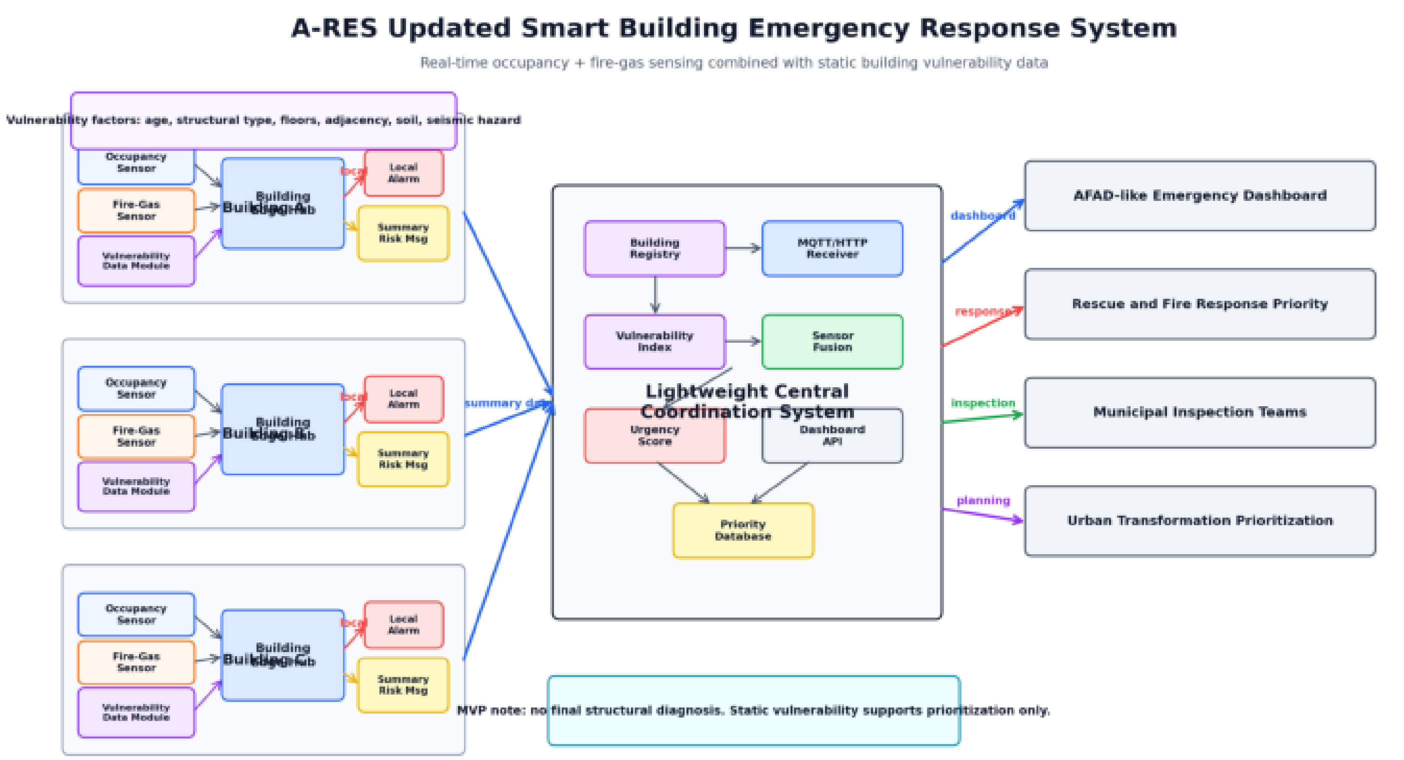
A-RES Updated System Design

Occupancy + Fire-Gas + Static Building Vulnerability Module



1. Updated System Decision

After the team discussion, the A-RES system scope has been revised to make the project more realistic, feasible, and academically defensible. The previous idea included a real-time structural health sensor placed on building columns. However, this component requires complex calibration, structural engineering validation, and real damage data. For the current MVP, this would increase technical risk and cost.

Therefore, the updated A-RES system removes the real-time structural health sensor from the MVP scope and replaces it with a **static building vulnerability module**.

The updated system has three main information sources:

```
1. Occupancy sensing
2. Fire-gas sensing
3. Static building vulnerability data
```

This keeps the project focused on post-earthquake emergency response prioritization while avoiding unsafe claims such as "the system can definitely determine whether a column is damaged."

2. Main System Purpose

A-RES is a post-earthquake smart building response system. Its main goal is to collect critical building-level data, calculate emergency priority, and support authorities such as AFAD-like emergency organizations, municipalities, fire departments, rescue teams, and inspection teams.

The system answers three practical questions:

```
1. How many people may still be inside the building?
2. Is there a secondary hazard such as fire, smoke, or gas leakage?
3. Is the building more vulnerable because of age, construction type, adjacency, soil, or local seismic hazard?
```

These answers are combined into an urgency score that helps prioritize response.

3. Why the Structural Health Sensor Was Removed from MVP

The idea of placing sensors on columns is technically valuable, but it is difficult to implement correctly in a short university project. A column monitoring system may require:

- Strain gauges or displacement sensors
- Sensor calibration for each building
- Structural engineering reference models
- Baseline measurements before the earthquake
- Real post-earthquake damage labels for validation
- Professional installation on structural elements

Because of these requirements, real-time structural health monitoring is better positioned as a future enhancement rather than an MVP component.

The safer report statement is:

Real-time structural health monitoring is excluded from the MVP scope due to calibration, validation, and installation complexity. Instead, A-RES uses static building vulnerability indicators such as building age, structural type, number of floors, adjacent-building condition, soil class, and local seismic hazard.

This decision makes the project more feasible and easier to defend.

4. Updated Architecture

The updated system has two real-time sensor groups and one static data module.

```
[Occupancy Sensor / Entrance Camera]
[Fire-Gas Sensor]
[Building Vulnerability Data Module]
          |
          v
[Building Edge Hub]
          |
          v
[Lightweight Central Coordination System]
          |
          v
[AFAD-like Dashboard / Emergency Teams / Municipal Decision Makers]
```

The edge hub still exists, but its role becomes more focused. It collects real-time sensor data, performs local warning checks, and sends summarized data to the central coordination system. Static building vulnerability data is stored in the central building registry and can also be cached locally on the edge hub.

5. Information Source 1: Occupancy Sensing

The occupancy sensing component estimates how many people may be inside the building. This is important because post-earthquake emergency response should prioritize buildings where more people may be at risk.

For the MVP, occupancy estimation is performed through entrance camera processing:

```
Entrance video
        -> YOLO11n person detection
        -> ByteTrack person tracking
        -> virtual line crossing
        -> entry / exit count
        -> estimated current occupancy
```

The system does not perform face recognition or identity detection. It only estimates anonymous occupancy counts.

Example output:


```
{
  "building_id": "B-001",
  "occupancy": 42,
  "occupancy_confidence": 0.91
}
```


6. Information Source 2: Fire-Gas Sensing

The fire-gas sensor group detects secondary hazards that may occur after an earthquake. Even if a building does not collapse, fire, smoke, gas leakage, or carbon monoxide can create immediate danger.

Possible sensors:

Sensor	Purpose
MQ-2	Smoke and flammable gas detection
MQ-7	Carbon monoxide detection
BME280 / DS18B20	Temperature and environmental anomaly support

Example output:


```
{
  "building_id": "B-001",
  "smoke_detected": true,
  "gas_detected": false,
  "temperature": 58.0
}
```


This data can trigger a local alarm and increase the central urgency score. It can also indicate whether fire department or hazardous material response may be required.

7. Information Source 3: Static Building Vulnerability Data

Instead of using a real-time structural health sensor, A-RES uses static building vulnerability factors. These factors help estimate whether a building is more likely to be risky after an earthquake.

Recommended vulnerability factors:

Factor	Why It Matters
Building age	Older buildings may have been built under weaker regulations
Structural type	Reinforced concrete, masonry, steel, etc. behave differently
Number of floors	Taller buildings may have different evacuation and structural risks
Adjacent-building condition	Adjacent buildings may transfer impact or limit evacuation/access
Soil class / microzonation	Weak soil may amplify shaking
Local seismic hazard	Some locations have higher expected ground motion
Building use	Hospital, school, residence, factory, and tower have different response needs

This module produces a `building_vulnerability_index` between 0 and 1.

Example:


```
{
  "building_id": "B-001",
  "building_age": 38,
  "structural_type": "reinforced_concrete",
  "floors": 8,
  "adjacency_type": "adjacent",
  "soil_risk": "high",
  "seismic_hazard": "medium",
```

```
  "building_vulnerability_index": 0.72
}
```

8. Where Building Vulnerability Data Can Be Obtained

The following sources can be used or referenced depending on availability.

8.1 Building Age

Possible sources:

- Building permit records
- Occupancy permit / usage permit records
- Municipality archives
- Property or parcel records
- Scenario-based assumptions for MVP

In a real deployment, the most reliable source would be official municipality records or building permit data. In the MVP, building age can be simulated or manually entered into the building registry.

8.2 Adjacent-Building Condition

Adjacent-building condition describes whether a building is detached, adjacent, or part of a block arrangement.

Possible sources:

- Zoning status documents
- Municipality zoning plan systems
- Parcel and cadastral maps
- Manual map inspection for MVP

Suggested categories:

```
detached
adjacent
block
unknown
```

Example scoring:

```
| Adjacency Type | Risk Score |
|---|---:|
| Detached | 0.2 |
| Block | 0.5 |
| Adjacent | 0.8 |
| Unknown | 0.6 |
```

8.3 Soil and Ground Conditions

Soil condition is important because weak or soft ground can amplify earthquake shaking.

Possible sources:

- Building-specific geotechnical investigation report
- Municipality microzonation maps
- AFAD earthquake hazard map
- MTA geological maps
- Scenario-based assumptions for MVP

Correct report framing:

For real deployment, building-specific geotechnical reports should be used whenever possible. For MVP and city-scale simulation, microzonation maps and official hazard maps can be used as reference data.

8.4 Local Seismic Hazard

Local seismic hazard can be estimated from official earthquake hazard maps. This does not replace a soil report, but it provides regional ground motion context.

Possible sources:

- AFAD Turkey Earthquake Hazard Map
- Local municipality earthquake risk studies
- Academic microzonation reports

9. Suggested Building Vulnerability Index

For the MVP, a simple weighted index is enough. The purpose is not to make a final engineering judgment, but to estimate relative vulnerability for emergency prioritization.

Suggested formula:

```
building_vulnerability =
0.25 * age_score
+ 0.20 * structural_type_score
+ 0.10 * floor_score
+ 0.15 * adjacency_score
+ 0.20 * soil_score
+ 0.10 * seismic_hazard_score
```

Each component is normalized between 0 and 1.

Example scoring:

Factor	Example	Score
Building age	38 years	0.75
Structural type	reinforced concrete	0.50
Floors	8 floors	0.60
Adjacency	adjacent	0.80
Soil	high risk	0.85
Seismic hazard	medium	0.50

The final result is a relative vulnerability score:

```
building_vulnerability_index = 0.69
```

10. Updated Urgency Score Logic

The central urgency score should combine real-time and static factors.

Recommended high-level formula:

```
urgency_score =
occupancy_risk
+ fire_gas_risk
+ building_vulnerability_risk
+ data_confidence_adjustment
```

Suggested weights:

Component	Suggested Weight
Occupancy risk	35%
Fire-gas risk	30%
Building vulnerability	25%
Data confidence / freshness	10%

This can be adjusted with the Industrial Engineering and Civil Engineering teams.

Priority levels:

Score Range	Priority
0-30	LOW
31-60	MEDIUM
61-80	HIGH
81-100	CRITICAL

11. Edge Hub Role in the Updated System

The edge hub remains important. It does not need to perform all final decisions, but it provides fast local reaction and reduces unnecessary data transmission.

Edge hub responsibilities:

1. Collect real-time occupancy and fire-gas data.
2. Perform basic threshold checks.
3. Trigger local alarms if fire/gas or critical conditions are detected.
4. Send summarized building status to the central system.
5. Buffer recent data if network connection is lost.
6. Use cached building vulnerability values if needed.

Example summarized edge message:

```
{
  "building_id": "B-001",
  "occupancy": 42,
  "fire_gas_risk": false,
  "local_alarm": "NORMAL",
  "building_vulnerability_index": 0.69,
  "timestamp": "2026-05-12T14:30:00Z"
}
```

12. Lightweight Central Coordination Role

The central system is not a large expensive data center. It can be a small server, university machine, cloud VPS, or laptop for the MVP.

Its responsibilities:

- Receive summarized data from buildings
- Store building registry and vulnerability information
- Calculate urgency scores
- Rank all buildings by priority
- Provide dashboard/API to AFAD-like authorities and emergency teams
- Provide optimization data to the Industrial Engineering team

The central system is necessary because each edge hub only knows its own building. The central system sees all buildings and can compare them.

13. Updated System Roadmap

Phase 1: MVP Demonstration

Task	Output
Build building registry with vulnerability fields	CSV / JSON building registry
Simulate occupancy and fire-gas sensor data	MQTT messages
Calculate vulnerability index	`building_vulnerability_index`
Calculate urgency score	LOW / MEDIUM / HIGH / CRITICAL
Show dashboard	Priority sorted building list

Phase 2: Validation

Task	Output
Phone doorway video test	Manual vs detected occupancy table
Fire-gas threshold simulation	Fire/gas event handling
Vulnerability scoring examples	5-10 building scenario table
Latency test	MQTT-to-dashboard timing

Phase 3: Report Integration

Task	Output
Explain static vulnerability module	Report section
Add cost/feasibility update	Financial and feasibility support
Add architecture illustration	Report figure
Provide IE optimization input	CSV dataset

Phase 4: Future Extension

Task	Output
Add real structural health monitoring	Future R&D
Integrate official municipality data	Real building registry
Add secure communication	TLS and device authentication
Add API integration for authorities	AFAD-like dashboard/API

14. Cost and Feasibility Impact

Removing the real-time structural health sensor from the MVP reduces cost and technical risk.

Previous Structural Sensor Challenge

Issue	Impact
Requires installation on columns	Hard to test in university setting
Requires calibration	High engineering complexity
Requires damage labels	Difficult to validate
Sensor cost can scale with number of columns	Expensive for real deployment

Updated Static Vulnerability Module

Advantage	Impact
Uses existing building information	Lower cost
Can be simulated for MVP	Easier demonstration
Helps Civil Engineering contribution	Better interdisciplinary fit
Works with IE optimization	Better data for resource allocation
Avoids unsafe structural diagnosis claims	More academically defensible

In short, the updated system is more feasible for the current project timeline.

15. Recommended Report Statement

The following text can be used directly in the report:

In the updated A-RES architecture, real-time structural health sensors are excluded from the MVP scope due to calibration, installation, and validation complexity. Instead, the system uses a static building vulnerability module based on building age, structural type, number of floors, adjacent-building condition, soil class, and local seismic hazard. This module is combined with real-time occupancy and fire-gas sensor data to calculate an urgency score. The purpose is not to provide a final structural safety diagnosis, but to support emergency response prioritization, evacuation decisions, inspection planning, and long-term urban transformation prioritization.

16. Visual Generation Prompt

```
Create a clean professional engineering diagram for "A-RES: Updated Smart Building Emergency Response System".

Show three smart buildings on the left. Inside each building, show two real-time sensor groups:
1. Occupancy Sensor / Entrance Camera
2. Fire-Gas Sensor

Also show a "Building Vulnerability Data Module" connected to each building. This module should include:
- building age
- structural type
- number of floors
- adjacent-building condition
- soil class / microzonation
- local seismic hazard

Connect the real-time sensors and the building vulnerability module to a "Building Edge Hub". The edge hub
should include:
- local data collection
- local alarm trigger
- summarized risk message
- MQTT/HTTP transmission

Show arrows from the edge hubs to a lightweight "Central Coordination System". The central system should
include:
- building registry
- vulnerability index calculation
- sensor fusion
- urgency score engine
- dashboard/API

On the right side, show AFAD-like emergency dashboard, rescue teams, fire teams, municipal inspection teams, and
 urban transformation prioritization.

Make clear that the structural health sensor has been removed from the MVP and replaced by static building
vulnerability data. Add a small note: "No final structural diagnosis; vulnerability data supports prioritization
 only."

Use a clean white background, blue and gray technical style, red only for alarms/critical priority, readable
labels, no cartoon style, no face recognition imagery, and no unnecessary decoration.
```

17. Conclusion

The updated A-RES system is more realistic and easier to defend. It uses:

```
Real-time occupancy sensing
+ Real-time fire-gas sensing
+ Static building vulnerability data
+ Edge hub preprocessing
+ Lightweight central coordination
```

This design still supports the main project goal: helping emergency authorities prioritize rescue, fire response, evacuation, inspection, and long-term urban transformation decisions after an earthquake.